

Maksim KLEIMENOV

Master's Student in High-Energy Physics
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Summary

M2 High-energy Physics student at IP Paris with aspirations in astroparticles and neutrino physics. Experienced in phenomenological modeling and statistical data analysis with Python and C++.

Education & Relevant Coursework

M.Sc. in High-Energy Physics

Institut Polytechnique de Paris (IP Paris)

GPA: 18.3/20.0

Ongoing (since Sept 2024)

Palaiseau, Île-de-France, France

B.Sc. in Applied Mathematics and Physics

Moscow Institute of Physics and Technology (MIPT)

GPA: 4.93/5.00

Sept 2020 – Jul 2024

Dolgoprudny, Moscow Region, Russia

High-energy Physics: QFT I-III, Particle physics, Neutrino physics, Electroweak interactions, QCD

Astrophysics: Stellar Astrophysics, Compact Objects, Astroparticles, Gen. Relativity, Cosmology, Grav. Waves

Plasma Physics: Space Plasmas, MHD, Kinetic Theory, Waves and Instabilities, Astrophysical plasmas

Research Experience

Extended emission around Pulsar Wind Nebulae

Laboratoire AstroParticule & Cosmologie (APC)

Independent Project

Ongoing (since Feb 2026)

Université Paris Cité, Paris, France

Advisers: Dr. Gwenael Giacinti and Dr. Dmitri Semikoz

Task: Explain extended X-ray and γ -ray emission misaligned with PWN tails

Skills: Phenomenological modeling and particle propagation in turbulent magnetic field

Searching for extreme blazars in H.E.S.S. data

Laboratoire Leprince-Ringuet (LLR)

M2 internship project

Ongoing (since Oct 2025)

IP Paris, Palaiseau, France

Adviser: Dr. Stephen Fegan (with Dr. Mathieu de Naurois and Dr. Alessandro Montanari)

Tasks: – Search for extreme blazars in H.E.S.S. data and determine their spectra and lightcurves

– Put constraints on EBL and IGMF based on the spectra and GeV halos

Skills: Fermi-LAT and H.E.S.S. data analysis (*fermipy*, *ParisAnalysis*, *gammapy*); SSC Blazar Models

Lepto-hadronic model of Galactic microquasar V4641 Sgr

Laboratoire AstroParticule & Cosmologie (APC)

M1 internship project (19/20)

Mar – Jul 2025

Adviser: Dr. Dmitri Semikoz (with Dr. Foteini Oikonomou and Dr. Andrii Neronov)

Tasks: – Develop lepto-hadronic model of the γ -ray signal (Inv. Compton, Bremsstr., Synchr., π -decay)

– Study the source morphology and anisotropic diffusion of cosmic rays in local GMF

Skills: – Phenomenological modeling (analytical and *am3*) with parametric optimization (MCMC)

– Analysis of galactic dust; CO, HI, and HII maps

Results: Reports at ICRC 2025 (F.O.) and ICAP 2026 (M.K.) and an article under review in PRD

Signatures of EBL absorption in StEVECat data

Laboratoire Leprince-Ringuet (LLR)

M1 research project (18/20)

Nov 2024 – Mar 2025

IP Paris, Palaiseau, France

Adviser: Dr. Stephen Fegan

Task: Develop a tool for EBL absorption calculation, analyse StEVECat entries and derive EBL impact

Skills: Curved space-time opacity (*astropy*, *gammapy*); Bayesian inference and MCMC (*emcee*)

Baikal-GVD sensitivity to selected neutrino sources

Laboratory of High Energy Neutrino Astrophysics

Bachelor thesis (10/10)

Nov 2022 – Jul 2024

Institute for Nuclear Research, Moscow, Russia

Adviser: Dr. Dmitry Zaborov

Tasks: Compute Earth propagation effects, calculate sensitivity for point sources and the Gal. Ridge

Skills: – Neutrino propagation (*ν FATE*), BDT (C++, *ROOT*, *TMVA*) and Feldman-Cousins method (*PyROOT*)

– Optimized numerical integrations (*numpy*, *scipy*, *numba*)

Results: Reports at AYSS 2024, MFTI 2025; publications in PEPAN Letters. Co-author in Baikal-GVD collab.

Skills

Programming: Python (astropy, gammapy, am3, pyROOT, fermipy); C++ (ROOT, TMVA, GSL); Kotlin

Data analysis: Likelihood fitting, Bayesian inference, MCMC

Languages: English (C1, IELTS: 8.0), French (B2), German (B1), Russian (native)

Publications

1. **M. Kleimenov**, A. Neronov, F. Oikonomou, and D. Semikoz. *Leptonic and Hadronic Models of High-energy Nebula Around V4641 Sgr*. Awaiting reference in PRD. [arXiv:2512.13578](https://arxiv.org/abs/2512.13578).
2. **M. Kleimenov**, A. Neronov, F. Oikonomou, and D. Semikoz. *Multimessenger emission from the PeVatron Microquasar V4641 Sagittarii*. PoS (ICRC 2025) 786. DOI: [10.22323/1.501.0786](https://doi.org/10.22323/1.501.0786).
3. **M. I. Kleimenov**, D. N. Zaborov. *Expected Neutrino Rates from Point-Like Astrophysical Sources in Baikal GVD*. Phys. Part. Nuclei Lett. 21, 646–649 (2024). DOI: [10.1134/S1547477124700912](https://doi.org/10.1134/S1547477124700912).
4. V. A. Allakhverdyan, A. D. Avrorin, A. V. Avrorin, **M. I. Kleimenov**, D. N. Zaborov et al. (Baikal-GVD Collaboration). *Real-time follow-up of multimessenger alerts at the Baikal-GVD telescope* J. Phys.: Conf. Ser. 2984 012023 (2025). DOI: [10.1088/1742-6596/2984/1/012023](https://doi.org/10.1088/1742-6596/2984/1/012023)
5. V. A. Allakhverdyan, A. D. Avrorin, A. V. Avrorin, **M. I. Kleimenov**, D. N. Zaborov et al. (Baikal-GVD Collaboration). *Probing the Galactic neutrino flux at neutrino energies above 200 TeV with the Baikal Gigaton Volume Detector*. Astrophys. J. 982 (2025) 73. DOI: [10.3847/1538-4357/adb630](https://doi.org/10.3847/1538-4357/adb630).
6. D. V. Zaborov, V. A. Allakhverdyan, A. D. Avrorin, **M. I. Kleimenov** et al. *Neutrino Astronomy at Lake Baikal*. Phys. Atom. Nuclei 88, 852–858 (2025). DOI: [10.1134/S1063778825601374](https://doi.org/10.1134/S1063778825601374)

Conferences and Schools

1. **Initiative in Cosmology and AstroParticle Physics seminar** (organized by J. Silk) Mar 2026
Models of extended high-energy emission from microquasar V4641 Sgr IAP, Paris, France
2. **The 66th MIPT Conference** Theses are published in the “Trudy MFTI” journal Apr 2024
Expected neutrino rates in Baikal-GVD telescope (Second-degree diploma) Dolgoprudny, Russia
3. **The XXVII International Scientific Conference of Young Scientists and Specialists** Oct 2023
Expected neutrino rates from point-like astrophysical sources in Baikal-GVD (Best report) Dubna, Russia
4. **Astroparticle Symposium 2025** Institut Pascal, Orsay, France, Nov 2025
5. **Astroparticle Symposium 2024** Institut Pascal, Orsay, France, Nov 2024
6. **Autumn School for Physics of Quark-gluon matter** JINR, Dubna, Russia, Nov 2023
7. **23rd Baikal School on Physics of Elementary Particles and Astrophysics** Irkutsk, Russia, July 2023
8. **School for Physics and Technique of Particle Accelerators** JINR, Dubna, Russia, July 2023

Teaching & Leadership

- Intern-Teacher, Moscow School 1543** Developed and taught an elective physics course 2021–2024
- Intern-Teacher, Letovo School** Prepared laboratory works and in-class exercises 2023–2024

Honors and Awards

- I am the professional** Winner in “megascience” experimental setups section Moscow, Russia, Jun 2024
- 15th International Physicists’ Tournament** 3rd place (IPT Team) Paris, France, Apr 2023
- 13th International Physicists’ Tournament** Winner (Team Russia) Warsaw, Poland (online), Jul 2021

Miscellaneous Research Projects

Physicists’ Tournament Projects

1. “Mendocino meter” (Adv: Dr. A. Svetlichnyi, 7.8/10 at IPT-2023) Sep – Jan 2023
2. “Ferromagnetic patterns” (Adv: Dr. G. Fedorov, 9.0/10.0 at RuPT-2022) Sep – Mar 2022
3. “Finger of death” (Adv: Dr. V. Vanovski, 9.0/10.0 at IPT-2021, best reporter) Jan – June 2021

MIPT Project course

4. Restoration of tunnel diode parameters (Adv: Dr. S. Zhabin, 10/10) Apr-May 2023
5. Investigation of Compton effect on X-ray photons (Adv: Dr. A. Svetlichnyi, 9/10) Oct 2022 - Jan 2023
6. Designing a setup for scanning interferometry (Adv: Dr. V. Vanovski, 10/10) Feb – May 2022
7. Updated DANNs sterile neutrino detector sensitivity (Adv: Dr. N. Skrobova, 9/10) Oct – Dec 2021
8. Cumulative jets in granular media (Adv: Dr. S. Vinogradov, 9/10) Oct – Dec 2020